

A working model of behavior leakage: a constant write, gated by context similarity

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1 Setup

Leakage: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appears under other, untrained contexts. We measure it as the trained–base change in the behavior’s expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander’s own response (#502/#489), judge-scored agreement rates for sycophancy (#509), misaligned-and-coherent rates for emergent misalignment (#521), taught-fact emission for fact implants (#500).

We summarize a context c by an activation vector $v_c \in \mathbb{R}^d$ and a behavior by the *behavior state* $v_b(c) = f(c)$, the activations where the behavior is expressed, with f the model’s unknown context-to-behavior map. Each behavior also has a *read-out direction* r_b whose projection tracks expression (the unembedding row for the marker; a probe or contrastive direction otherwise). The constructions of v_c and v_b (layer, position, aggregation) are swept, not fixed; the experiments in §4 search over them.

2 The simplest model

Assumptions. Each comes with the relaxation that applies when it fails; the formula survives every relaxation in a weaker form.

A1 (behaviors are linear directions).

Adding a single residual-stream direction elicits a behavior [3, 4]; projections onto single directions track traits before generation [5]; emergent misalignment is predicted, controlled, and ablated through one direction [6, 7]. *Relaxation.* The *write* need not copy the behavior direction: it can be a router, a query shift that recruits pre-existing attention circuitry which fetches the behavior from the context’s own values [8]. The sharpest evidence is a rank-1 EM adapter whose B vector has cosine 0.04 to the misalignment direction at its own layer while 98% of its effect ablates along that direction downstream [7]. The behavior stays one direction at read-out, but prediction P1 below can fail by mechanism when fetched content differs across contexts.

A2 (edits are read/write pairs).

Weight updates read input directions and write output directions, with the rank-one outer product as the atomic unit [1]; low-rank adapters make this explicit [16, 2]. Narrow fine-tunes act as a near-constant steering write [8, 9], and the parameterization is irrelevant: at matched implant strength, LoRA and full fine-tuning leak identically (gap +0.00 nat, #514). *Extension.* The edit need not come from training: a transformer block provably converts an ICL prefix into a transient rank-one weight patch [17, 18], so an in-context prefix

performs the same write $\times$ gate operation for the duration of the prompt, and the gate already ranks leakage from ICL sources (#489). Full ICL=fine-tuning equivalence remains open for pretrained models [19, 20, 21], but in-context behavior carries predictive signal about fine-tuning outcomes: ICL recovers $> 85\%$ of fine-tuning’s corrections in-distribution [19], in-context inferences anticipate and repair fine-tuning’s generalization failures [22], and demonstration effects compress to a single latent direction [23]. Hence prediction P7 below.

A3 (linearity, the extra assumption).

At one layer, $v_b \approx Mv_c$, and the edit acts at that layer. *Relaxation.* For a small update, the change at any input is first-order: $\Delta f(c) \approx K(c, c') \times \text{residual}$, with K the empirical tangent kernel [14], a regime measured to hold for language-model fine-tuning in several settings [15]. The formula survives with $g(c) = K(c, c')/K(c', c')$, where K is the cross-kernel between the read-out gradient at c and the training-loss gradient at c' (symmetric only for read-out-aligned losses such as marker-only). The dot product of equation (1) is the linear-kernel case, and the practical ladder runs cosine $\rightarrow$ distributional comparisons (Gaussian-KL, MMD) $\rightarrow$ the exact cross-kernel via backprop; gains over cosine have so far been marginal (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy). The gate can also stop being a similarity altogether: under the routing picture of A1’s relaxation it is attention-shaped, near-binary for discrete triggers, and estimable from samples rather than computable from cosines.

A4 (minimal edit).

Training implements the smallest weight change that fits the new pair: the rank-one key $\rightarrow$ value write used by linear associative memories and ROME [10, 11, 12]. *Relaxation.* The write becomes a low-rank subspace, $\Delta v_b(c) = U \text{diag}(s) V^\top v_c$; prediction P1 below weakens to “shifts span r dimensions,” and the SVD already in use measures r . With multiple training pairs the edit solves a kernel system over all pairs; contrastive negatives enter as off-diagonal terms that subtract the common mode (persona contexts sit near cosine 0.92, which is why positive-only implants leak near-uniformly while contrastive ones localize: $\sim 99.6\%$ source vs $\sim 11.7\%$ bystander, #247/#329). Early stopping spectrally filters the solution toward raw similarity, so the time-indexed predictor interpolates between raw and Gram-corrected similarity; neither endpoint has been run.

From assumptions to formula. Fine-tune context c' from behavior state $v_{b'} = f(c')$ to an achieved state $v_{b''}$ (under early stopping the achieved state undershoots the target, so the write is measured, not assumed). A4 with A3 gives the edit: minimizing $\|\Delta M\|_F$ subject to $(M + \Delta M) v_{c'} = v_{b''}$ yields

$$M' = M + \frac{(v_{b''} - v_{b'}) v_{c'}^\top}{\|v_{c'}\|^2} \quad \implies \quad \Delta v_b(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: what leaks}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{gate: who leaks}}. \quad (1)$$

Each assumption does one job: A4 makes the edit rank-one (one write direction, one key), A3 makes the gate the activation inner product, A2 makes the same structure visible in the weights ($\Delta W \approx \text{write} \otimes \text{key}$), and A1 makes the write measurable, since the observed change is one slope per behavior times the gate: $\Delta z_b(c) = \langle r_b, v_{b''} - v_{b'} \rangle g(c)$. The softmax compresses log-probabilities near saturation and can reorder contexts there, which is why we store logits alongside log-probabilities.

3 Predictions

P0 (weights decompose).

The top singular vectors of the trained ΔW are the key and the write. Under contrastive training the key moves from raw $v_{c'}$ toward a source-minus-negatives contrast direction as dose grows.

P1 (one write direction).

Per-bystander activation shifts at the read-out all point along $(v_{b''} - v_{b'})$; an SVD of the shift matrix is near rank-one with a top component matching the behavior delta.

P2 (the gate is base-model context similarity).

Some fixed kernel on base-model context summaries sets who leaks, and it transfers across held-out panels. On the linear rung it is the dot product, hence asymmetric: $c' \rightarrow c$ scales as $\cos \theta \cdot \|v_c\|/\|v_{c'}\|$.

P3 (the behavior’s prior matters, through three channels).

Source side: the write scales with the residual $\|v_{b''} - v_{b'}\|$, so a source that already performs the behavior produces little leakage despite high similarity. Bystander side: the prior enters through read-out compression (measurement) and possibly through the gate itself (mechanism); the two separate in logit space, but only with the exact gate rather than a proxy.

P4 (additivity).

Two implants trained jointly leak like the sum of their singletons, with deviations set by the sources’ mutual similarity (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).

The gate predicts best at the layers where the behavior is computed; scoring this needs an independent localization (patching or probing), since scoring by predictor quality is circular.

P6 (cross-behavior transfer).

The same write moves other behaviors in proportion to its projection onto their read-out directions, extending the rule from context generalization to behavior generalization.

P7 (ICL proxy).

The base model’s in-context behavior on the training mix forward-predicts the post-training leakage map (see A2’s extension for why).

No single miss refutes the model, since each assumption carries a relaxation (§2); the conjunction does. With the gate computed exactly and the write measured on a rig-clean implant, a failure of write $\times$ gate to rank leakage on a held-out panel leaves no version standing.

4 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]’s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction ($\cos 0.96\text{--}0.98$); the marker implant does not, and its trained context is <i>least</i> aligned. Benign SFT also gives one direction (#552, running); positive-only retrain tests the rig confound (#561, running).
P2	Correlate base-model similarity with measured leakage across source×bystander panels (#502, #489, #509)	Similarity ranks marker leakage ($ \rho \approx 0.76\text{--}0.79$, layers 19–24) and extends to ICL/multi-turn contexts, but only ordinally (out-of-fold $R^2 \approx 0$, single-seed); structure does not transfer across panels (#553); the fact arm reversed sign ($ \rho = 0.49$) while the base prior predicted correctly (+0.27, #500), an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander’s own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); the measurement-vs-mechanism split awaits the exact gate.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker predicts best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as a leakage predictor	Queued; the similarity gate already extends to ICL source contexts (#489).

The open headline experiment is a dose titration: implant at a non-saturated anchor (#448), measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432→#456), at several doses across the 10–25-step cliff (#139), with at least 3 seeds. At low dose the shift matrix should be near rank-one with similarity-graded magnitude; across the cliff the direction should rotate and the prediction collapse. The P1 split already complicates the low-dose half, so the titration doubles as its diagnosis.

What cannot be relaxed. Every version assumes the update is small in weight/feature space, though not in behavior: read-outs amplify, so a lazy edit can flip emission discretely or induce EM through nine rank-1 adapters [7], and tripling implant depth left the edit geometry unchanged (#538). If training moves the features themselves (the rich regime), no base-model predictor can

work, and the historical EM predictor failure looks like that regime ($\rho = 0.09$ prose-only #458; assistant-axis rotation $38\text{--}53^\circ$, discrimination collapse, #125). The current tension: the marker arm passes the gate test but fails the one-direction test, while EM does the reverse; #552 and #561 decide whether the rig or the regime explains it. The durable question under all versions is the same: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

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